MERN STACK

TRAINING TR-103 REPORT DAY 37

6 August 2026

Overview:

Day 37 of the MERN Stack training focused on developing the Notification module for the CareerConnect – Career Guidance and Management System. I learned how to create a notification interface that allows users to view important updates related to job applications and other activities within the job portal. The module retrieves notification records from the backend using Express.js REST APIs and displays them dynamically through React.js components. MongoDB is used to store and manage notification information associated with individual users. The module also supports notification read status management, helping users distinguish between new and previously viewed updates. This development improved my understanding of API integration, database operations, React hooks, and user-specific notification management.

Learning Objectives:

- To understand the development of a Notification module using the MERN Stack.
- To learn how to create, retrieve, and update notification records in MongoDB.
- To understand REST API development using Node.js and Express.js.
- To use the `useState()` and `useEffect()` hooks for managing and fetching notification data.
- To implement read and unread notification status functionality.
- To understand how notifications can be associated with individual users.
- To learn how backend events can be connected with notification generation.
- To improve knowledge of frontend-backend integration and dynamic data rendering.

Project: CarrerConnect – Carrer Guidance and Management System

NOTIFICATION MODULE DEVELOPMENT

- The Notification module is developed to keep users informed about important activities and updates in the CareerConnect system.
- It provides a centralized interface where users can view notifications related to job applications, application status updates, and other relevant job portal activities.
- The module is implemented using React.js for the frontend and Express.js with MongoDB for backend processing.
- The `useState()` hook is used to manage notification records, unread notification counts, and loading messages.
- The `useEffect()` hook is used to fetch notification data automatically when the Notification page is opened.
- The module communicates with the backend through REST APIs to retrieve notifications, mark notifications as read, and manage notification records.
- Notifications are associated with the relevant user so that each user can access their own updates.
- The module helps users stay informed about important activities without repeatedly checking individual sections of the job portal.

NOTIFICATION API DEVELOPMENT

The Notification APIs are developed using Express.js to retrieve and manage notification records stored in MongoDB.

1. Get Notifications API

GET `/notifications/:userId`

- Retrieves notification records associated with a specific user.
- Fetches notification information from MongoDB.
- Returns notification details, including the notification message, date, type, and read status.
- Displays notifications related to job applications and other supported system activities.

- Sends the notification data to the frontend for display.

2. Create Notification API

POST `/notifications`

- Creates a new notification for a specific user.
- Receives the user ID, notification message, notification type, and other required information.
- Validates the notification data before storing it.
- Saves the notification record in MongoDB.
- Allows the system to inform users about relevant events, such as application status changes, when this functionality is integrated.

3. Mark Notification as Read API

PUT `/notifications/:id/read`

- Updates the read status of a selected notification.
- Identifies the notification using its unique ID.
- Changes the notification status from unread to read.
- Updates the notification record in MongoDB.
- Refreshes the notification display and unread count after a successful update.

FEATURES OF THE NOTIFICATION MODULE

- Displays notifications related to job applications and other important activities.
- Shows notification messages with their corresponding dates and times.
- Differentiates between read and unread notifications.
- Displays the total number of unread notifications, if implemented.
- Allows users to mark notifications as read.
- Retrieves user-specific notification records from MongoDB.
- Updates the notification list after relevant actions.
- Provides appropriate messages when notifications are unavailable.

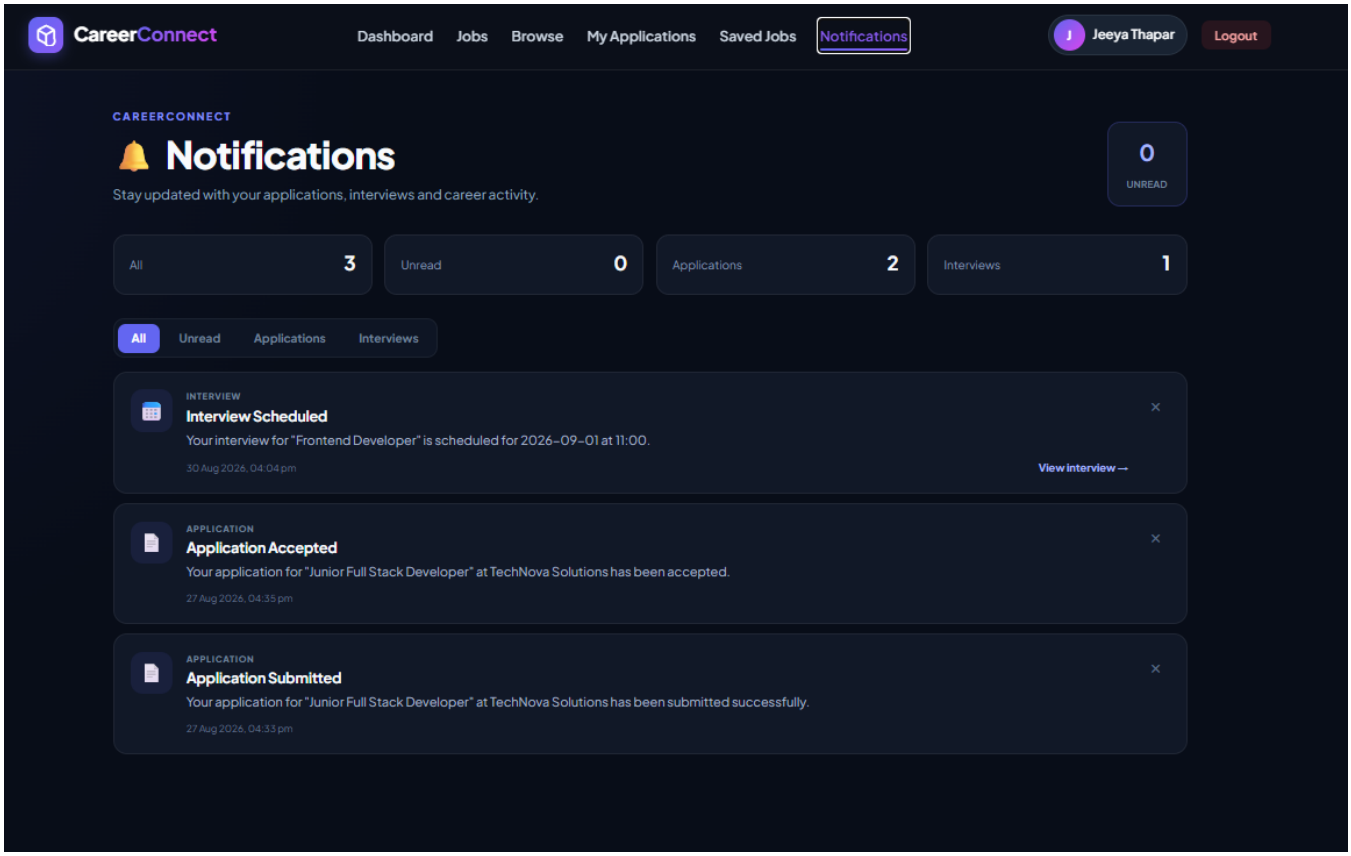
- Improves communication between the system and its users.

WORKING OF THE NOTIFICATION MODULE

- The user logs in to the CareerConnect – Career Guidance and Management System.
- The user opens the Notification page from the navigation menu.
- The Notification component retrieves notification records using the Get Notifications API.
- The backend fetches notifications associated with the logged-in user from MongoDB.
- The API returns the notification data to the frontend.
- The received notification records are stored using the `useState()` hook.
- The notifications are displayed with their messages, dates, and read or unread status.
- When a relevant system event occurs, such as an application status update, a notification can be created through the Create Notification API.
- When the user opens or reads a notification, the Mark Notification as Read API updates its status.
- The frontend refreshes the notification list and unread count, if available.
- Users can view their updates in one place without navigating through multiple sections of the application.
- The Notification module improves the overall user experience by providing an organized way to access important job portal updates.

Conclusion

The Notification module provides an organized way for users to receive and view important updates within the CareerConnect system. It helps users stay informed about job application activities and other relevant events through a centralized notification interface. The integration of React.js, Express.js, Node.js, and MongoDB enables the retrieval and management of notification records. Through the development of this module, I gained practical knowledge of REST API integration, database operations, React hooks, and notification status management. Overall, the module improves user engagement and makes the job portal more informative and user-friendly.


CareerConnect
Dashboard
Jobs
Browse
My Applications
Saved Jobs
Notifications
Jeeya Thapar
Logout
CAREERCONNECT
Notifications
Stay updated with your applications, interviews and career activity.
0
UNREAD
All 3
Unread 0
Applications 2
Interviews 1
All
Unread
Applications
Interviews
INTERVIEW
Interview Scheduled
Your interview for "Frontend Developer" is scheduled for 2026-09-01 at 11:00.
30 Aug 2026, 04:04 pm
View interview →
APPLICATION
Application Accepted
Your application for "Junior Full Stack Developer" at TechNova Solutions has been accepted.
27 Aug 2026, 04:35 pm
APPLICATION
Application Submitted
Your application for "Junior Full Stack Developer" at TechNova Solutions has been submitted successfully.
27 Aug 2026, 04:33 pm